Tea Sense — Project Proposal

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Tea Sense will test whether a portable NIR spectrometer plus a chemometric / machine-learning model can estimate the moisture content of black tea in seconds, against oven-drying reference values, under practical testing conditions.

Field	Detail
Project name	Tea Sense: A Spectroscopic System for Estimating Moisture and Quality Parameters in Black Tea
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Programme	Final Year Project (FYP)
Date	October 2026
Version	1.0 (proposal draft built from the PPRS document and literature survey matrix)
Keywords	Near-Infrared Spectroscopy (NIR), tea moisture, chemometrics, machine learning, spectral preprocessing, quality control

0. Research Progression and Coverage

Each later decision in this proposal traces back to an earlier one: tools serve the methodology, methods serve objectives, and objectives answer the evidenced problem.

Stage of the research progression	Where it is covered
Idea → researchable topic	Section 0 (idea screening, title evolution, idea-to-topic canvas)
Problem	Section 2 (problem statement built from context, limitation, consequence and research need)
Evidence	Section 2 (evidence and impact tables)

Stage of the research progression	Where it is covered
Literature	Section 4 (search log, screening, survey matrix, thematic synthesis)
Gap	Section 4 (gap statements and gap evidence test)
Aim / objectives	Section 3 (aim, objectives, scope, alignment map)
Requirements	Section 6
Proposed solution	Section 5
Tools / alternatives	Section 7
Methodology / methods	Section 8 (methodology family choice, Research Onion, methods, objective–method–evidence matrix)
Evaluation	Section 9
LESP / EDI / sustainability	Section 10 (BCS Code of Conduct mapping, EDI, SDGs)
Schedule / risks	Sections 11 (WBS, timeline) and 12 (risk register)
Ethics application and supervisor review	Section 16
PP slides	Section 17
Live defence	Section 18

Idea screening

The starting point was a problem area (slow moisture testing in tea), not a preferred tool. The idea was screened before committing.

Screening question	Answer for Tea Sense
Problem significance	Moisture affects shelf life, safety and the sale weight of about 245,787 t of Sri Lankan tea exports (2025); reference testing takes hours
Researchability	Produces measurable evidence: prediction error against an oven reference, effect sizes of preprocessing and models, field timing

Screening question	Answer for Tea Sense
Computing relevance	Core challenge is signal processing, feature selection, machine learning and validation design on high-dimensional spectra
Literature base	28 relevant sources retained, including 7 on NIR moisture prediction
Feasibility	Portable NIR, oven and balance are obtainable; open-source software; sample count scaled to one academic year
Evaluation	Clear ground truth (oven-drying) and standard metrics (RMSEP, R^2, RPD, bias)
Professional responsibility	Risks are misuse as a certified result, supplier confidentiality and lab safety; all can be managed
Scope	Narrowed to one parameter (moisture), one tea type (black) and one instrument class

From idea to working title

Level	Tea Sense	What changed
Idea	Measuring tea quality with sensors	Very broad; many parameters and technologies
Area	Spectroscopy and machine learning for food and agricultural quality	Defines the technical domain
Topic	NIR spectroscopy with chemometric and ML models for moisture estimation in black tea	Fixes the sensor, the target parameter and the product
Working title	Tea Sense: A Spectroscopic System for Estimating Moisture and Quality Parameters in Black Tea	States what will be estimated and in what product

How the literature changed the topic:

Point in process	Focus
Before literature	Detect many tea quality and safety parameters (moisture, polyphenols, glyphosate, heavy metals) with one device
After reading Revilla et al. (2024) and Qiu et al. (2026)	Dropped trace contaminants: no direct NIR absorption and weak validation for metals
After reading the moisture studies (Zou 2022, Lan 2022, Tao 2026)	Focused on moisture, where water bands give a direct NIR signal
After noticing single-site validation everywhere	Added held-out and source-held-out testing plus a field feasibility test
Recommended refined title	Evaluating NIR Spectroscopy and Machine-Learning Pipelines for Rapid Moisture Estimation in Sri Lankan Black Tea

The refined title is offered for the supervisor to consider: it matches the actual scope (moisture only) more closely than "moisture and quality parameters". It avoids words such as "novel" or "first" that the results have not yet earned.

Idea-to-topic canvas

Prompt	Response
Area of interest	Rapid, non-destructive food quality testing
Observed computing problem	Turning noisy, high-dimensional NIR spectra into reliable moisture estimates that generalise beyond one lab setting
Who or what is affected	Tea factories, QC staff, brokers, exporters and laboratories
Evidence the problem exists	Slow, destructive oven method; NIR models in the literature validated on single cultivars or sites
Why this is a computing problem	Requires preprocessing, variable selection, regression modelling, leakage-free validation and a software prototype
Existing approaches known so far	Oven-drying, moisture analysers, PLSR / SVR / deep models on NIR spectra

Prompt	Response
Possible limitation or gap	No Sri Lankan black-tea model; weak external validation; feasibility rarely measured
Data and resources available	Portable NIR (to confirm), lab oven and balance, tea samples from suppliers or retail
How success could be evaluated	RMSEP, R^2p, RPD and bias on locked and source-held-out data; time per sample; field accuracy
Initial ethical, legal and social risks	Misuse of predictions, supplier confidentiality, lab safety, small personal-data footprint
Working topic or title	See refined title above

1. Project Overview

Tea Sense will design, build and evaluate a NIR spectroscopy workflow that predicts black-tea moisture content in under a minute per sample, and will report honestly how close it gets to the oven-drying reference.

Working title: Tea Sense: A Spectroscopic System for Estimating Moisture and Quality Parameters in Black Tea.

Project summary: Moisture is a critical quality parameter for tea. It governs shelf life, mould risk, storage behaviour, processing efficiency (especially at the final drying stage) and market value. The standard way to measure it is gravimetric: weigh a sample, dry it in an oven to constant mass, and weigh it again. That method is accurate but slow, destructive and tied to a laboratory.

Near-Infrared (NIR) spectroscopy measures how a sample reflects light between roughly 780 and 2,500 nm. Water has strong O-H absorption bands in this region (around 970, 1,450 and 1,940 nm), so NIR spectra carry information about moisture. A spectrum takes seconds to record and leaves the sample intact.

This project will:

1. Collect NIR spectra from a set of black-tea samples covering a realistic moisture range.
2. Measure the true moisture of each sample by oven-drying to constant mass.
3. Compare spectral preprocessing methods (raw, SNV, MSC, Savitzky-Golay smoothing and derivatives).
4. Compare wavelength-selection methods (full spectrum versus CARS and related methods).

5. Compare predictive models (PLSR as the chemometric baseline, then SVR, Random Forest and a small neural network where the data supports them).
6. Evaluate the best pipelines on held-out samples using RMSEP, R^2, RPD and bias.
7. Run a small practical / field feasibility test outside the controlled laboratory set-up.
8. Package the selected model in a simple software prototype that turns a new spectrum into a moisture estimate.

Target users or stakeholders:

Stakeholder	Interest in Tea Sense
Tea factory managers and dryer operators	Fast moisture checks at the final drying and sorting stages to avoid under- or over-drying
Factory and estate quality-control staff	A preliminary screening tool that reduces how many samples need full laboratory testing
Tea brokers, exporters and packers	Rapid checks on incoming and outgoing lots before storage, blending or shipment
Tea research bodies (e.g. Tea Research Institute of Sri Lanka)	Evidence on whether low-cost NIR is viable for Sri Lankan black tea
Laboratory analysts	A complementary method that keeps oven-drying as the reference, not a replacement
Academic community (IIT / University of Westminster)	A reproducible comparison of preprocessing, wavelength selection and models for tea moisture
The researcher	Skills in spectroscopy, chemometrics, ML evaluation and research methodology

Expected outcome: A documented, reproducible pipeline and dataset that answers four research questions:

- **RQ1:** To what extent can NIR spectra predict tea moisture compared with oven-drying reference values?
- **RQ2:** How do SNV and Savitzky-Golay preprocessing affect prediction performance?
- **RQ3:** How do predictive models and wavelength selection (e.g. CARS) affect performance?
- **RQ4:** To what extent is the approach feasible for rapid measurement under practical or field conditions?

The final output is an evidence-based conclusion on feasibility, with clear limits, not a claim that NIR can replace laboratory testing.

2. Problem Statement and Context

The tea industry needs moisture values faster than oven-drying can give them, and NIR has not yet been validated for that job on Sri Lankan black tea under practical conditions.

Problem statement: There is a need to evaluate whether spectroscopy-based sensing, combined with chemometric and AI-based modelling, can provide a sufficiently accurate and rapid method for estimating moisture content in tea under practical testing conditions.

The statement is built from evidence, not opinion:

Element	Question	Tea Sense
Context	What is happening and in what setting?	Tea factories and buyers check moisture repeatedly; the accepted method is gravimetric oven-drying in a laboratory
Limitation / challenge	What is difficult or inadequate?	Oven-drying is slow, destructive and lab-bound; published NIR moisture models are validated on single cultivars, sites and seasons (Zou et al., 2022; Lan et al., 2022; Tao et al., 2026), and none use Sri Lankan black tea
Consequence	Why does it matter?	Drying decisions are made before results arrive; generalisation of NIR models to local tea is unknown, so they cannot be trusted for screening (Qiu et al., 2026)
Research need	What needs investigation?	A controlled, reproducible comparison of NIR preprocessing, wavelength selection and models for Sri Lankan black-tea moisture, tested on held-out samples and in practical conditions

Background and context

Tea is a major agricultural and commercial product. Its quality depends on physical, chemical and safety parameters, including moisture, polyphenols, caffeine, pigments (theaflavins, thearubigins), pesticide residues and heavy metals. Measuring these accurately supports processing, storage, grading, safety and pricing decisions.

Sri Lanka is a major global tea producer. In 2025, Sri Lankan exports across bulk, packeted, tea-bag, instant and green tea categories totalled about 245,787 metric tonnes. At that scale, even small improvements in how quickly quality is checked can affect large volumes of product.

Black-tea manufacture runs through withering, rolling, fermentation (oxidation), drying and sorting. Moisture falls from around 75–80% in fresh leaf to a target of roughly 2.5–4% in made tea. If made tea is under-dried, it risks mould and loss of quality in storage. If it is over-dried, it can taste burnt, lose aroma and lose saleable weight. Moisture is therefore checked repeatedly through production and again before packing and export.

Current challenge or limitation

- **Reference methods are slow.** Gravimetric oven-drying needs repeated heating and weighing until mass is constant, which takes from tens of minutes to several hours per batch.
- **Reference methods are destructive.** The tested sample is consumed and cannot be sold or re-tested.
- **Reference methods need a laboratory.** Samples are carried away from the line, so results arrive after the processing decision has already been made.
- **Other quality parameters are harder still.** Polyphenols need extraction and Folin–Ciocalteu spectrophotometry; gallic acid and sugars need HPLC; metals need ICP-MS. These are accurate but costly and slow.
- **Human tasting is subjective.** Sensory grading is essential for pricing but varies between tasters.
- **NIR models do not travel well.** Published NIR models often lose accuracy when moved to new cultivars, regions, seasons or instruments (Qiu et al., 2026).

Evidence supporting the problem

- Hazarika et al. (2017) describe conventional tea-quality chemistry as invasive, skill-intensive, time-consuming and dependent on costly, sometimes hazardous chemicals, and present NIR as a green alternative.

- Qiu et al. (2026) review ten spectroscopic technologies across agricultural products and list poor model universality, environmental interference, equipment cost and data volume as unresolved challenges.
- Revilla et al. (2024) report 98.3% calibration accuracy but lower validation accuracy (92.3%) on some tea blend categories, showing that calibration figures overstate real performance.
- Moisture-specific NIR studies exist but are narrow: Zou et al. (2022) used one cultivar (Jiukeng) under controlled processing; Lan et al. (2022) used one cultivar at one site during green-tea fixation; Andasuryani et al. (2024) used Gambir herbal tea, not *Camellia sinensis*; Tao et al. (2026) used one white-tea production line in one season.
- None of the reviewed moisture studies used Sri Lankan (Ceylon) black tea or tested a model across samples from several estates.

Impact of the problem

Area	Impact
Product quality	Under-dried tea risks mould and quality loss in storage; over-dried tea loses aroma and briskness
Commercial value	Tea is sold by weight, so moisture errors change revenue and can lead to rejected lots
Process efficiency	Late results mean dryers cannot be adjusted while a batch is still running
Laboratory load	Every check needs staff time, oven capacity and destroyed sample
Food safety and compliance	Moisture limits apply in export and buyer specifications (e.g. ISO 3720 for black tea), and excess moisture raises microbial risk

Opportunity to address

Portable NIR spectrometers in the 900–1700 nm range now cost a fraction of benchtop instruments and can connect to a laptop or phone. Open-source tools (Python, scikit-learn) make it practical to build and compare chemometric and machine-learning models. This creates an opportunity to build a rapid, non-destructive screening method that:

- returns a moisture estimate in seconds rather than hours;
- keeps the sample intact;
- can be used near the production line;

- flags samples that need full laboratory confirmation.

If moisture estimation works, the same pipeline can later be tested on other parameters (for example polyphenols), each validated on its own merits.

3. Aim, Objectives and Scope

The project has one aim, seven measurable objectives, and a deliberately narrow scope: moisture in black tea first, everything else later.

Overall aim: To design, develop and evaluate a NIR spectroscopy-based approach for the rapid estimation of moisture content in tea using spectral preprocessing, wavelength selection and predictive modelling.

Objective	Expected outcome	Success measure
O1. Review and critically evaluate research on NIR spectroscopy, chemometrics and machine learning for tea quality and moisture analysis	Literature review chapter, literature survey matrix (30+ sources) and research gap matrix	Each method chosen in O3–O5 is justified by at least two sources; gaps are mapped to RQs
O2. Collect NIR spectra and matching oven-drying reference moisture values from a selected range of tea samples	Documented dataset of samples, spectra and reference values with a data dictionary	Target of at least 120 samples (minimum 80), moisture spread across the expected range, triplicate reference drying with repeatability SD ⩽ 0.2%
O3. Investigate the effect of spectral preprocessing (SNV, Savitzky-Golay, MSC, derivatives) on moisture prediction	Comparison table of preprocessing combinations under one fixed model and validation scheme	Cross-validated RMSECV and R^2 reported for every combination; best combination identified with justification
O4. Investigate wavelength selection (CARS, plus a comparison method such as SPA or interval PLS)	Selected wavelength subsets and their link to known water absorption bands	Performance of selected subset is within or better than full-spectrum performance using fewer variables

Objective	Expected outcome	Success measure
O5. Develop and compare predictive models (PLSR baseline, SVR, Random Forest, and an ANN/1D-CNN if data allows)	Trained, tuned and saved models with documented hyperparameters	All models evaluated on the same splits; best model statistically compared with PLSR baseline
O6. Evaluate models on held-out samples using RMSEP, R^2, RPD and bias	Final evaluation report on the untouched test set	Target $R^2p \geq 0.90$, RPD ≥ 2.5 and RMSEP small enough for screening (aim ≤ 0.5% moisture on made tea)
O7. Assess practical feasibility of rapid moisture estimation under selected field or operational conditions	Field feasibility report: time per measurement, robustness, usability, observed limitations	Time per sample under 60 seconds; field RMSEP and bias reported against oven reference for a field subset

Why each objective exists

Objective	Why it exists	Research question served
O1	Establishes current knowledge and shows the gap is real, not assumed	All
O2	No public dataset of Sri Lankan black-tea spectra with oven moisture exists; the study must create one	RQ1–RQ4
O3	Preprocessing strongly changes NIR results (Das et al., 2020), and the best choice for moisture is unknown	RQ2
O4	Wavelength selection can match full-spectrum accuracy with fewer variables (Hazarika et al., 2024), which matters for low-cost devices	RQ3
O5	It is unclear whether non-linear models beat PLSR on small tea datasets	RQ3
O6	Calibration figures overstate accuracy (Revilla et al., 2024); only held-out results count	RQ1

Objective	Why it exists	Research question served
O7	Accuracy alone does not prove practical value; speed and robustness must be measured	RQ4

End-to-end alignment map

Gap	Research question	Objective	Key requirements	Method	Evaluation evidence
G1 No validated moisture model for Sri Lankan black tea	RQ1	O2, O6	FR01–FR05, FR13, NFR01	Paired NIR + oven dataset; locked test set	RMSEP, R^2p, RPD, bias vs oven
G2 Effect of preprocessing for moisture unknown	RQ2	O3	FR06, FR07	Factorial comparison with fixed PLSR and CV	RMSECV per preprocessing; ablation
G2 Effect of wavelength selection unknown	RQ3	O4	FR09	CARS vs SPA/iPLS vs full spectrum	RMSECV, variable count, band stability
G3 Calibration does not guarantee generalisation	RQ3	O5, O6	FR08, FR10–FR12	Same splits for all models; leave-one-source-out	RMSEP per model and per held-out source; paired tests
G4 Practical feasibility untested	RQ4	O7	FR15–FR18, NFR02, NFR03, NFR06	Prototype in field; oven-checked subset	Time per sample, repeatability, field RMSEP, SUS

In scope

- NIR spectroscopy as the primary sensing technology (portable spectrometer, 900–1700 nm, diffuse reflectance).
- Black tea (*Camellia sinensis*) samples, mainly made tea and late-stage drying samples, from selected Sri Lankan sources.
- Moisture content as the primary and only guaranteed target parameter.
- Oven-drying to constant mass as the reference (ground-truth) method.
- Spectral preprocessing: raw, SNV, MSC, Savitzky-Golay smoothing, first and second derivatives, and combinations.
- Wavelength selection: CARS, with at least one comparison method.
- Predictive models: PLSR (baseline), SVR, Random Forest, and optionally an ANN or 1D-CNN.
- Evaluation on held-out samples with standard chemometric metrics.

- A limited practical / field feasibility test.
- A simple software prototype (Python script or small app) that loads a spectrum and returns a predicted moisture value.

Out of scope

- Guaranteed measurement of polyphenols, caffeine, theaflavins, glyphosate, pesticide residues or heavy metals. These are named as possible future extensions only.
- Designing or building custom spectrometer hardware.
- Commercial certification, regulatory approval or claims that the system replaces laboratory testing.
- Full in-line integration into a factory dryer or conveyor.
- Calibration transfer between several different spectrometer models.
- Green, white, oolong or herbal teas, except where used as an optional robustness check.
- Sensory grading or price prediction.

Assumptions and constraints

Assumptions

- Oven-drying gives sufficiently reliable reference moisture values when done in triplicate under a fixed protocol.
- Collected samples will span enough moisture variation for meaningful regression (achieved by sourcing from different stages and by controlled conditioning if needed).
- Access to a portable NIR spectrometer will be available for the data-collection period.
- Access to a laboratory oven and an analytical balance (0.001 g) will be available.
- Water absorption features in 900–1700 nm (around 970, 1,200 and 1,450 nm) are strong enough to carry the moisture signal.

Constraints

- **Time:** one academic year, with fixed university submission deadlines.
- **Budget:** student-level; no purchase of high-end benchtop FT-NIR instruments.
- **Samples:** dependent on the goodwill of estates, factories or brokers; numbers and diversity will be limited.
- **Equipment:** one spectrometer model; results may not transfer to other instruments.
- **Environment:** field conditions (humidity, light, temperature) cannot be fully controlled.
- **Generalisability:** findings apply to the cultivars, regions, processing stages and instrument tested.

4. Background Research and Existing Solutions

The literature shows NIR already predicts tea moisture well in single-cultivar, controlled settings; the open question is whether it holds up on Sri Lankan black tea across sources and outside the lab.

Research questions guiding the review

- What tea-quality parameters has NIR spectroscopy been used to predict, and how well?
- Which spectral ranges, instruments and measurement modes (reflectance, transmittance) are used for tea moisture?
- Which preprocessing methods (SNV, MSC, Savitzky-Golay, derivatives) most improve tea and moisture models?
- Which wavelength-selection methods (CARS, Random Frog, SPA, VISSA, mRMR) help, and by how much?
- How do linear chemometric models (PLSR) compare with machine-learning and deep-learning models?
- How are models validated, and how often is independent or external validation reported?
- What limits transfer from the laboratory to the factory or field?
- What alternative spectroscopic technologies (UV-Vis, FTIR, THz, LIBS, hyperspectral imaging) exist, and why is NIR preferred for moisture?

Sources and search approach

- **Databases:** Google Scholar, ScienceDirect, MDPI, SpringerLink, Scopus, IEEE Xplore, Nature/Scientific Reports, IOP Science and Sri Lankan symposium proceedings (ISAE, RESCON).
- **Search strings:** ("near infrared" OR NIR OR NIRS) AND tea AND (moisture OR "water content"); "tea quality" AND spectroscopy AND (chemometrics OR "machine learning"); CARS AND PLSR AND tea; "portable NIR" AND tea; "black tea" AND drying AND spectroscopy.
- **Date window:** mainly 2016–2026, with older foundational sources where needed.
- **Inclusion:** peer-reviewed articles, reviews or conference papers on spectroscopy for tea, food or agricultural quality, with quantitative results or a clear methodological contribution.
- **Exclusion:** non-food NIR (medical oximetry, neuroscience), studies with no quantitative evaluation, and duplicates.

- **Screening result:** 33 records were logged in the literature survey matrix. Five were excluded as off-domain (gemology, two cerebral-oximetry papers, a neuroscience commentary and an RFID antenna paper). 28 were retained, of which seven deal directly with moisture.
- **Recording:** every source is logged in the survey matrix with summary, gap, contribution, algorithms, tools, dataset, limitations, future work and citation (Appendix C).

Search strings built from concept blocks (synonyms joined with OR, blocks joined with AND):

Concept block	Terms
Product / domain	tea OR "Camellia sinensis" OR "black tea" OR "CTC tea" OR "made tea"
Target parameter	moisture OR "water content" OR "moisture content" OR drying OR withering
Sensing technology	"near infrared" OR NIR OR NIRS OR "Vis-NIR" OR "portable spectrometer" OR "micro-NIR"
Computing / modelling	chemometrics OR PLSR OR "partial least squares" OR "machine learning" OR SVR OR "deep learning" OR CARS
Evaluation concern	validation OR RMSEP OR generalisation OR robustness OR "external validation"

Example: (tea OR "Camellia sinensis") AND (moisture OR "water content") AND ("near infrared" OR NIR) AND (PLSR OR "machine learning" OR chemometrics).

Screening layers:

1. **Title screening:** remove clearly irrelevant results (e.g. medical NIRS).
2. **Abstract screening:** check problem, sensor, target and data relevance.
3. **Full-text screening:** confirm usable methods and quantitative results.
4. **Quality check:** sample size, reference method, validation design, reported limitations.
5. **Backward snowballing:** references of Zou et al. (2022) and Qiu et al. (2026).
6. **Forward snowballing:** newer papers citing Zou et al. (2022) and Lan et al. (2022), which surfaced Tao et al. (2026) and Kang et al. (2026).

The search log (database, date, string, hits, kept) is kept in Appendix A; the matrix records for each paper the problem, data, preprocessing, method, evaluation, findings, limitations and relevance.

Existing solutions: key sources

Source / existing solution	Key findings	Limitations	Relevance
Zou et al. (2022), moisture of black tea during processing with a miniaturised NIR (Horticulturae)	305 spectra across fresh leaf, withering, rolling, fermentation and drying; SNV/MSC/SG/Z-score compared; GWO-tuned SVR outperformed PLS	Single cultivar (Jiukeng), controlled conditions; no multi-factory test	Closest match: same target, same 900–1700 nm portable class, oven reference; sets the benchmark design
Tao et al. (2026), deep learning for real-time moisture in white-tea withering (Scientific Reports)	17,856 line observations; MS-580 NIR moisture analyser above a conveyor; MSC + mRMR + attention BiGRU; chronological train/val/test split	One region and season; white tea, not black	Shows real-line deployment and leakage-safe chronological splitting
Kang et al. (2026), TSNet with sigma-adaptive spectral sampling for tea moisture (J. Food Meas. Charact.)	215 batches, 1,505 spectra, 0–80% moisture; GRU + self-attention + multi-scale CNN; online validation	Lab calibration with deliberately over-dried samples; transfer unverified	Shows deep models need large, wide-range datasets; useful upper reference
Lan et al. (2022), WOA-Elman moisture prediction in green-tea fixation (Foods)	Miniature NIR (NIR-S-R2) + image features; SNV; PLSR vs SVR vs Elman vs WOA-Elman; fusion helped	One cultivar, one site, one run	Confirms low-cost micro-NIR can track moisture; supports PLSR vs non-linear comparison
Andasuryani et al. (2024), moisture in Gambir herbal tea by NIR (IOP Conf.)	60 samples, 40/20 calibration/validation, PLS, oven reference, 1000–2500 nm	Herbal tea, one variety, small set	Simple template for a PLS + oven-reference study at small sample size
An et al. (2025), withering quality of black tea by multi-information fusion (Foods)	IAS-3100 NIR + machine vision; VISSA selection; SVR; fusion levels compared; moisture analyser reference	Single cultivar and controlled withering	Supports wavelength selection and fusion as future extension

Source / existing solution	Key findings	Limitations	Relevance
Huang et al. (2022), THz + NIR fusion for moisture in cattle feed (Frontiers in Physics)	Derivative + SG preprocessing; CARS feature fusion; PLSR; 2:1 calibration/validation	Not tea; THz needs pellets and dry air	Demonstrates CARS + PLSR for moisture specifically
Hazarika et al. (2024), CTC black tea quality and market valuation by portable NIR	100 CTC samples; SNV, MSC, 1D, 2D; PLSR with CARS and Random Frog; few wavelengths gave strong models	Reference labels partly sensory; moisture not the target	Black tea, portable 900–1700 nm, CARS + PLSR: methodological template
Revilla et al. (2024), NIRS for heavy metals and blend percentage in tea (Foods)	322 samples; MPLS, PCA, SIMCA, OPLS-DA; 98.3% calibration vs 92.3% validation accuracy	Metals have no direct NIR peak; small sub-categories	Evidence that validation must be independent; caution on indirect predictions
Das et al. (2020), green-tea discrimination by gallic acid using NIR	PCA + PLSR with HPLC reference; preprocessing raised separability index from 53.63 to 93.20	Five green-tea variants; gallic acid, not moisture	Quantifies the effect of preprocessing
Hazarika et al. (2017), NIR as a green tool for tea quality (review)	NIR (780–2500 nm) diffuse reflectance with chemometrics as a reagent-free alternative	Review; no validated moisture model	Foundational justification for NIR in tea
Qiu et al. (2026), spectroscopic detection in agricultural quality (review)	Surveys 10 technologies; SG/SNV/MSC; SPA/CARS/GA; PLSR, SVM, ANN, CNN	Cross-product review	Lists the open challenges this project tests: universality, environment, cost
Li et al. (2023), Keemun black tea grading by data fusion (Food Chem X)	Smartphone micro-NIR + image + colorimetric array; SVM, LS-SVM, ELM	One supplier; grinding required	Shows phone-linked NIR is practical for black tea

Source / existing solution	Key findings	Limitations	Relevance
Zhang, Gao and Guo (2026), Shiqian Taicha quality 2021–2023 (Applied Sciences)	116 commercial batches; FT-NIR predicted moisture and other constituents	Commercial, not controlled; climate confounding	Shows NIR moisture used in routine multi-year monitoring
Zhou et al. (2026), tea variety traceability by NIR (Beverages)	SG, MSC, SNV, 1D; PCA, LDA, SPA, SVM	Four varieties, 84 spectra	Supports preprocessing choices; warns about small sets
Damayanthi et al. (2025, 2026), sugar adulteration in Sri Lankan black tea by FTIR and UV-Vis	Seven Sri Lankan regions; SNV, MSC, PCA, OPLS-DA, PLS-DA, RBF-SVM	Abstracts only; limited validation detail	Local precedent for chemometrics on Ceylon tea
Piyasena et al. (2022, 2026), Sri Lankan tea sugars and polyphenols	HPLC and Folin–Ciocalteu reference chemistry at St. Coombs Estate	Chemical, not spectroscopic	Background on local reference methods and future polyphenol extension
Huang et al. (2024), THz identification of coffee roast and variety (Foods)	THz-TDS with PCA, LD, SVM, KNN	THz blocked by water; needs moisture < 5%	Shows why THz is unsuitable as the primary moisture sensor

Key themes and insights

1. **NIR works for tea moisture in controlled settings.** Every moisture study reviewed reports usable models, because water has strong NIR absorption.
2. **Preprocessing matters a lot.** SNV, MSC, SG and derivatives appear in almost every study; Das et al. (2020) show separability rising from 53.63 to 93.20.
3. **Wavelength selection helps.** CARS, Random Frog, VISSA and mRMR reduce variables and often improve or match full-spectrum results.
4. **PLSR is the standard baseline.** Non-linear models (SVR, Elman, BiGRU, CNN) sometimes beat it, but mainly with larger or more varied datasets.
5. **Portable 900–1700 nm devices are now common.** NIR-S-R2, NIR-S-G1, IAS-3100 and similar handheld units appear repeatedly.

6. **Validation is usually weak.** Most studies use one cultivar, one site, one season and a random split; few test on new sources.
7. **Fusion is a growing trend.** NIR plus images or other sensors improves accuracy but adds cost and complexity.

Thematic synthesis and implications

Theme	What the studies agree or disagree on	Implication for Tea Sense
Data and sample characteristics	Datasets range from 60 samples (Andasuryani 2024) to 17,856 line observations (Tao 2026); most come from one cultivar and site	Collect from several sources; record source metadata; report limits honestly
Reference measurements	Oven-drying or moisture analysers are standard for moisture; sensory labels add noise (Hazarika 2024)	Use triplicate oven-drying with a written SOP
Preprocessing	Nearly universal use of SNV/MSC/SG; gains are large but vary by task (Das 2020)	Compare preprocessing systematically rather than pick one
Wavelength selection	CARS, Random Frog, VISSA and mRMR keep or improve accuracy with fewer variables	Test CARS against a second method and full spectrum
Linear vs non-linear models	PLSR is the baseline everywhere; SVR, Elman and deep models win mostly with larger datasets (Zou 2022, Lan 2022, Kang 2026)	PLSR baseline, SVR and RF as candidates; deep models only if data allows
Validation design	Mostly random splits; few external tests; calibration-validation drops reported (Revilla 2024)	Lock test set; add leave-one-source-out; report RMSEC alongside RMSEP
Portability and deployment	Micro-NIR and phone-linked units work (Li 2023, Lan 2022); in-line systems exist (Tao 2026)	Use a portable unit; measure time and field accuracy

Theme	What the studies agree or disagree on	Implication for Tea Sense
Alternative technologies	THz blocked by water (Huang 2024); LIBS destructive; FTIR/UV-Vis used for adulteration, not moisture	NIR remains the right primary sensor for moisture

Existing and similar solutions

Solution type	Example	Strength	Limitation for this problem
Oven-drying (gravimetric)	ISO 1573-style method	Accurate reference	Slow, destructive, lab-bound
Halogen / infrared moisture analysers	Bench moisture balances	Faster than oven (minutes)	Still destructive; one sample at a time; one parameter only
Capacitive / resistive moisture meters	Handheld grain/tea meters	Instant and cheap	Sensitive to packing and temperature; poor at low moisture; no spectral data for other parameters
Commercial in-line NIR analysers	Conveyor-mounted NIR moisture units (e.g. MS-580 in Tao 2026)	Continuous, non-destructive	Costly; calibrations proprietary and tied to one product and line
Research NIR models	Zou 2022, Lan 2022, Kang 2026	Demonstrate feasibility	Single cultivar or site; not Sri Lankan black tea; feasibility not measured

Identified gap or opportunity

- No reviewed study builds a moisture model for **Sri Lankan (Ceylon) black tea**.
- Moisture models are almost all **single-cultivar and single-site**; performance on samples from **different estates or grades** is not reported.
- Few studies compare **preprocessing × wavelength selection × model** systematically on the **same held-out split**.

- Few studies report **practical feasibility** (time per measurement, handling, environmental sensitivity) alongside accuracy.

Tea Sense addresses these by building and openly comparing pipelines on Sri Lankan black tea, holding out test samples (including a source-held-out check where possible), and running a field feasibility test.

Gap evidence test

Test question	Answer
Which papers support the gap?	Zou 2022, Lan 2022, Andasuryani 2024, Tao 2026, Kang 2026 (single-site moisture work); Revilla 2024 and Qiu 2026 (validation and universality problems)
Is it repeated or important?	Single-cultivar, single-site validation appears in every moisture study reviewed; Qiu (2026) names universality as a key bottleneck
Is it within computing?	Yes: preprocessing, feature selection, model comparison and validation design are the core work
Can this project investigate it?	Yes, at small scale: one instrument, several sources, one season
Can it be evaluated?	Yes: held-out and source-held-out RMSEP against an oven reference

The gap is stated as a need to *evaluate*, not as "nobody has done NIR on tea". NIR on tea is well established; what is missing is evidence for this tea, this sensor class and practical use.

Decisions informed by the literature

Decision	Informed by
Target moisture first, not trace contaminants	Revilla 2024; Qiu 2026
Use 900–1700 nm portable NIR	Zou 2022; Lan 2022; Hazarika 2024; Li 2023
Oven-drying as reference	Zou 2022; Andasuryani 2024
Compare SNV, MSC, SG and derivatives	Das 2020; Hazarika 2024; Zhou 2026

Decision	Informed by
Use CARS plus a second selection method	Hazarika 2024; Huang 2022; Qiu 2026
PLSR baseline with SVR and RF candidates	Zou 2022; Lan 2022; Qiu 2026
Lock a test set and test across sources	Revilla 2024; Qiu 2026; Tao 2026 (chronological split)
Measure field feasibility	Qiu 2026; Tao 2026

5. Proposed Solution

Tea Sense is a measure-preprocess-select-predict pipeline: a portable NIR scan goes in, a moisture estimate with a confidence flag comes out, and oven-drying stays the reference that every prediction is judged against.

Solution overview

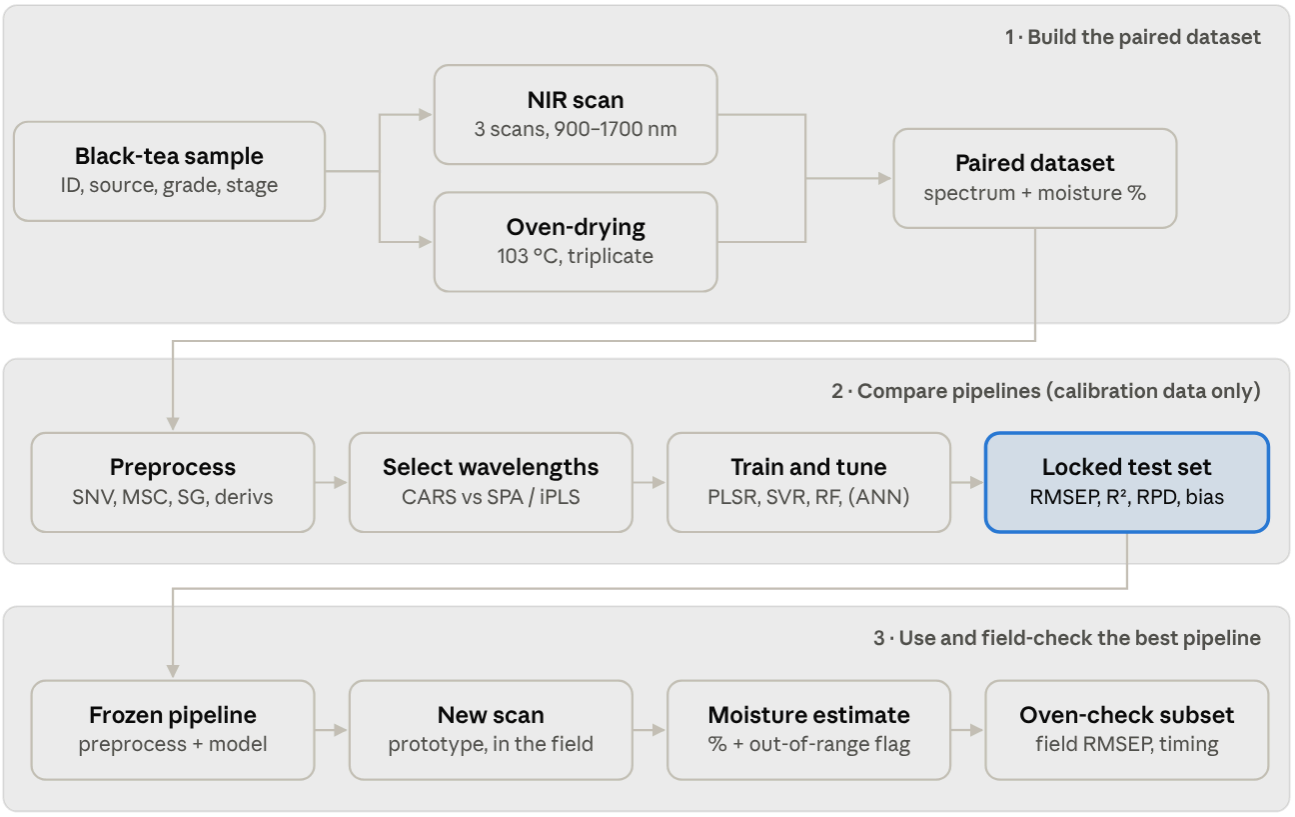
The solution has two halves. The **research half** builds a paired dataset (NIR spectrum + oven moisture for each sample) and runs a controlled comparison of preprocessing, wavelength selection and models. The **prototype half** freezes the best pipeline into a small tool that an operator can use: scan the sample, get a moisture value and a flag if the spectrum looks unlike the training data.

Core features or components

1. **Sample management.** Each sample gets an ID and metadata: source estate/factory, grade, processing stage, date, ambient temperature and humidity.
2. **Spectral acquisition.** Portable NIR spectrometer (900–1700 nm, diffuse reflectance) with dark and white-reference calibration before each session. Each sample is scanned three times after repacking, and spectra are averaged.
3. **Reference measurement.** Oven-drying to constant mass immediately after scanning, in triplicate, using a fixed temperature and time protocol (see Section 8).
4. **Data store.** Spectra and reference values saved as versioned CSV/Parquet files with a data dictionary.
5. **Preprocessing module.** Raw, SNV, MSC, Savitzky-Golay smoothing, first and second derivatives, mean-centring, and combinations.
6. **Outlier and quality control.** PCA scores with Hotelling's T^2 and Q-residuals to flag bad scans; checks on reference repeatability.

7. **Wavelength-selection module.** CARS, with SPA or interval PLS for comparison, and VIP scores for interpretation.
8. **Modelling module.** PLSR baseline; SVR, Random Forest and an optional ANN/1D-CNN; hyperparameter tuning by cross-validation on the training set only.
9. **Evaluation module.** RMSEC, RMSECV, RMSEP, R^2, RPD, bias, prediction-versus-reference plots and residual analysis.
10. **Prediction prototype.** A Python command-line tool or simple web/desktop interface that loads a new spectrum, applies the saved pipeline and returns moisture (%) plus an "outside model range" warning.

Every model is judged once on a locked test set before it reaches the field



Tea Sense pipeline · 3 stages, 12 steps

Each sample is scanned and oven-dried to form one paired record; pipelines are compared on calibration data only, and only the winner is tested once on locked data and then in the field.

High-level architecture / workflow

1. Sample is collected and logged.
2. Spectrometer is calibrated (dark + white reference).
3. Sample is scanned three times; spectra are averaged.

4. The same sample is oven-dried to give reference moisture.
5. Spectrum and reference are joined in the dataset.
6. Dataset is split into calibration and held-out test sets (Kennard-Stone or source-stratified; test set locked).
7. Preprocessing and wavelength selection are fitted on calibration data only.
8. Models are trained and tuned by cross-validation.
9. The best pipeline is evaluated once on the locked test set.
10. The frozen pipeline is deployed in the prototype and tried in the field against a small oven-checked subset.

Unique value

- **A moisture model for Sri Lankan black tea** in the reviewed literature.
- **Systematic comparison** of preprocessing × wavelength selection × model on one locked test set, instead of one hand-picked pipeline.
- **Honest generalisation test**, using held-out samples and, where possible, a held-out source.
- **Feasibility measured, not assumed:** time per scan, sensitivity to packing and ambient conditions, and usability are reported alongside accuracy.
- **Low cost and reproducible:** portable hardware and open-source Python, with published code and protocol.
- **Extensible:** the same pipeline can later be re-trained for polyphenols or other parameters.

Dependencies

Dependency	Why it is needed	Status / fallback
Portable NIR spectrometer (900–1700 nm)	Core sensor	To be confirmed with supervisor / lab; fallback is a university or partner benchtop NIR
Laboratory oven and analytical balance (0.001 g)	Reference moisture	University chemistry/food lab; fallback is a partner factory lab
Black-tea samples from several sources	Dataset variety	Estates, factories, brokers or retail; fallback is controlled re-conditioning of fewer lots

Dependency	Why it is needed	Status / fallback
Desiccator, sealed sample bags, humidity/temperature logger	Prevent moisture change between scan and drying	Low-cost purchase
Python 3, NumPy, SciPy, pandas, scikit-learn, matplotlib; optionally PyTorch	Analysis and modelling	Free and open source
Git/GitHub and Google Drive / OneDrive	Version control and backup	Available
Supervisor and domain-expert input (e.g. tea technologist)	Protocol review and interpretation	To be arranged

6. Requirements

The requirements cover the experiment (data and reference quality) as much as the software, because the model can only be as good as the paired data behind it. Priorities use MoSCoW (Must, Should, Could, Won't).

Functional requirements

ID	Requirement	Type	Priority	Rationale
FR01	Record each sample with a unique ID and metadata (source, grade, processing stage, date, ambient temperature and humidity)	Functional (data)	Must	Traceability; enables stratified splits and error analysis by source
FR02	Acquire NIR reflectance spectra over the instrument range (about 900–1700 nm) with dark and white-reference correction	Functional (acquisition)	Must	Core input; references remove lamp and detector drift
FR03	Take at least three scans per sample with repacking and store the averaged and individual spectra	Functional (acquisition)	Must	Reduces packing and particle-size effects; allows repeatability checks

ID	Requirement	Type	Priority	Rationale
FR04	Record oven-drying reference moisture (wet basis, %) in triplicate for every scanned sample	Functional (reference)	Must	Ground truth for training and evaluation
FR05	Join spectra and reference values in a single versioned dataset with a data dictionary	Functional (data)	Must	Reproducibility and auditability
FR06	Apply configurable preprocessing: SNV, MSC, Savitzky-Golay smoothing, first/second derivative, mean-centring and combinations	Functional (processing)	Must	Answers RQ2
FR07	Detect spectral and reference outliers using PCA (Hotelling's T^2, Q-residuals) and reference repeatability	Functional (QC)	Must	Prevents bad data from distorting models
FR08	Split data into calibration and locked test sets (Kennard-Stone and/or source-stratified)	Functional (evaluation)	Must	Answers Gap 3; prevents optimistic results
FR09	Perform wavelength selection with CARS and at least one comparison method (SPA or iPLS)	Functional (processing)	Must	Answers RQ3 and O4
FR10	Train and tune a PLSR baseline using cross-validation to pick latent variables	Functional (modelling)	Must	Standard chemometric baseline

ID	Requirement	Type	Priority	Rationale
FR11	Train and tune SVR and Random Forest models on the same splits	Functional (modelling)	Must	Answers RQ3; tests non-linear relationships
FR12	Train an ANN or 1D-CNN if the dataset is large enough (about 150+ samples)	Functional (modelling)	Could	Recent literature uses deep models; only justified with enough data
FR13	Compute RMSEC, RMSECV, RMSEP, R^2c, R^2cv, R^2p, RPD and bias for every pipeline	Functional (evaluation)	Must	Answers RQ1 and O6
FR14	Produce predicted-vs-reference plots, residual plots and regression-coefficient / VIP plots	Functional (output)	Should	Interpretation and examiner evidence
FR15	Save the selected pipeline (preprocessing + selected wavelengths + model) as a single loadable artefact	Functional (deployment)	Must	Needed for the prototype and field test
FR16	Prototype accepts a new spectrum file and returns predicted moisture (%)	Functional (prototype)	Must	Demonstrates rapid use (RQ4)
FR17	Prototype warns when a spectrum lies outside the calibration space (high T^2/Q)	Functional (prototype)	Should	Responsible use; avoids silent extrapolation
FR18	Prototype logs each prediction with timestamp, sample ID and ambient conditions	Functional (prototype)	Should	Field feasibility analysis
FR19	Simple graphical interface (web or desktop) for the prototype	Functional (prototype)	Could	Usability for non-programmers

ID	Requirement	Type	Priority	Rationale
FR20	Prediction of other parameters (polyphenols, caffeine)	Functional	Won't (this phase)	Out of scope; future extension

Non-functional requirements

ID	Requirement	Type	Priority	Rationale
NFR01	Prediction accuracy on the locked test set: target $R^2p \geq 0.90$ and RPD ⩾ 2.5	Performance (accuracy)	Must	Common chemometric thresholds for screening-grade models
NFR02	Measurement-to-result time under 60 seconds per sample, including three scans	Performance (speed)	Must	Core value against hours for oven-drying
NFR03	Repeatability: SD of predictions for repeated scans of one sample ⩽ 0.2% moisture	Reliability	Should	Operator trust; separates instrument noise from model error
NFR04	Every experiment reproducible from code, config and data version (fixed random seeds)	Reproducibility	Must	Academic integrity and examiner verification
NFR05	Prototype runs on a standard laptop without a GPU	Portability	Must	Field use; low cost
NFR06	Prototype usable after a 10-minute briefing (target SUS score ⩾ 68 from test users)	Usability	Should	Field feasibility (RQ4)
NFR07	Raw data backed up in two locations; no personal data stored	Security / data management	Must	Data-loss risk; privacy

ID	Requirement	Type	Priority	Rationale
NFR08	Supplier names anonymised in published outputs unless consent is given	Confidentiality	Must	Commercial sensitivity of partners
NFR09	Code documented and organised into modules with a README	Maintainability	Should	Hand-over and future extension
NFR10	Predictions shown with clear wording that they are preliminary and not a laboratory result	Transparency / safety	Must	Prevents misuse in commercial decisions

7. Tools and Technologies

NIR in the 900–1700 nm range, oven-drying, and an open-source Python stack give the best balance of moisture sensitivity, cost and reproducibility for a student project.

Need	Options considered	Selection criteria	Selected approach	Justification
Sensing technology	NIR spectroscopy; UV-Vis; FTIR-ATR; Raman; Terahertz (THz-TDS); LIBS; hyperspectral imaging; capacitive/resistive moisture meters	Sensitivity to water; speed; non-destructive; portability; cost; support in tea literature	**NIR diffuse reflectance spectroscopy**	Water O-H bands dominate NIR; strongest tea-moisture evidence (Zou 2022, Lan 2022, Tao 2026); THz is blocked by water; Raman is insensitive to water; LIBS is destructive; HSI is costly; simple meters give no spectral information for future parameters

Need	Options considered	Selection criteria	Selected approach	Justification
Spectrometer	Benchtop FT-NIR (Bruker MPA, Thermo Antaris II, Foss 5000, Buchi N-500); portable InGaAs units (NIR-S-G1, NIR-S-R2, IAS-3100, DLP NIRscan Nano class)	Range covering water bands; cost; availability; field portability; data export	**Portable InGaAs NIR spectrometer, about 900–1700 nm** (exact model to be confirmed); benchtop unit as fallback	Covers 970 / 1,200 / 1,450 nm water bands; used successfully for tea moisture (Zou 2022, Lan 2022); enables the field test (RQ4)
Reference moisture method	Oven-drying to constant mass; Karl Fischer titration; halogen/IR moisture analyser	Accuracy; accepted standard; cost; availability	**Oven-drying to constant mass** (ISO 1573-style: 103 ± 2 °C), triplicate	Accepted reference for tea; used in Zou (2022) and Andasuryani (2024); Karl Fischer is costlier and needs reagents; halogen analysers are useful as a quick cross-check only
Preprocessing	Raw; SNV; MSC; Savitzky-Golay smoothing; 1st/2nd derivative; detrending; EMSC	Proven in tea and moisture literature; simplicity; interpretability	**SNV, MSC, SG smoothing, SG 1st/2nd derivative and combinations**, compared systematically	Used in nearly all reviewed studies (Hazarika 2024, Zou 2022, Qiu 2026); directly answers RQ2
Wavelength selection	CARS; Random Frog; SPA; iPLS / siPLS; GA; VISSA; mRMR; VIP	Track record in tea; availability in Python; computational cost	**CARS (primary), SPA or iPLS (comparison), VIP (interpretation)**	CARS shown effective for black tea (Hazarika 2024) and moisture (Huang 2022); answers RQ3 / O4

Need	Options considered	Selection criteria	Selected approach	Justification
Predictive models	PLSR; MLR; SVR / LS-SVR; Random Forest; XGBoost; ELM; ANN; 1D-CNN; BiGRU/attention	Accuracy on small datasets; interpretability; risk of overfitting; literature support	**PLSR baseline; SVR; Random Forest; optional ANN / 1D-CNN**	PLSR is the chemometric standard; SVR beat PLS in Zou (2022); deep models need large data (Tao 2026, Kang 2026) so are optional
Programming and libraries	Python (scikit-learn, SciPy, NumPy, pandas, PyTorch); MATLAB + PLS_Toolbox; R (pls, prospectr); The Unscrambler	Free licence; reproducibility; flexibility; researcher skills	**Python 3 with scikit-learn, SciPy, NumPy, pandas, matplotlib; PyTorch if a neural model is built**	Open source and free; full control of pipelines and validation; MATLAB / Unscrambler need paid licences
Experiment tracking and versioning	Git + GitHub; MLflow; DVC; spreadsheets	Simplicity; reproducibility	**Git/GitHub for code; MLflow or structured CSV logs for runs; DVC optional for data**	Lets every reported number be traced to code, config and data version
Prototype interface	CLI script; Streamlit / Flask web app; desktop GUI; mobile app	Development effort; field usability; laptop compatibility	**Python CLI first, Streamlit web app if time allows**	Fast to build; runs on a field laptop; meets FR16–FR19
Documentation and project management	MS Word / Google Docs; Zotero / Mendeley; Trello / Notion; MS Project / GanttProject	Collaboration with supervisor; citation management	**Google Docs / Word, Zotero (Harvard style), Trello, Gantt chart**	Standard and free; supports supervisor review

8. Methodology and Implementation Plan

This is a quantitative, experimental study: build a paired dataset, compare pipelines under one fixed validation protocol, then test the winner once on locked data and once in the field.

Overall approach

Methodology, methods, development process and evaluation design are kept separate:

Term	Meaning	Tea Sense
Methodology	Overall logic for producing valid evidence	Quantitative experimental study, with an artefact (pipeline + prototype) built to carry out the investigation
Methods	Specific techniques inside that logic	Structured literature review, gravimetric reference, NIR acquisition, SNV/MSC/SG, CARS, PLSR/SVR/RF, cross-validation, paired significance tests, field timing, SUS
Development process	How the artefact is built	Iterative CRISP-DM-style cycle: understand data → prepare → baseline → candidates → tune → evaluate → package
Evaluation design	How claims are tested	Pre-specified metrics on a locked test set and source-held-out data, against PLSR and naïve baselines

Why this methodology family:

Candidate family	Fits when	Decision
Quantitative experimental	The central claim is measurable technical performance under controlled conditions	**Selected (primary):** RQ1–RQ3 are about prediction error under controlled factors
Design Science Research	The contribution is an artefact judged against objectives	**Partly used:** the prototype is an instrument for RQ4, not the main contribution
Case study	Context is essential and evidence comes from many sources	Rejected: the question is not about an organisation's practice
Qualitative study	The question is about meaning or experience	Rejected: user perceptions are not a research question
Survey	Attitudes across many participants are needed	Rejected: no valid need for a large sample of people

Candidate family	Fits when	Decision
Mixed methods	Technical metrics alone cannot answer the objectives	Light use only: a short SUS questionnaire supports RQ4 but does not drive conclusions

Research methodology (Saunders' Research Onion):

Layer	Choice	Reason
Philosophy	Positivism (with pragmatism for the prototype)	Moisture is objectively measurable; claims are tested against reference data
Approach	Deductive	Starts from literature-based expectations (e.g. preprocessing improves accuracy) and tests them
Methodological choice	Mono-method quantitative	Numerical spectra and reference values; statistical evaluation
Strategy	Experiment	Controlled comparison of preprocessing, selection and model factors
Time horizon	Cross-sectional (single collection period)	Fits the academic year; seasonal effects noted as a limitation
Techniques and procedures	Instrument measurement, gravimetric reference, statistical modelling	Standard practice in NIR chemometrics

Hypotheses:

- **H1 (RQ1):** NIR-based models predict tea moisture on held-out samples with $R^2p \geq 0.90$ and RPD ⩾ 2.5.
- **H2 (RQ2):** Scatter-corrected and smoothed spectra (SNV or MSC with SG) give lower RMSECV than raw spectra.
- **H3 (RQ3a):** CARS-selected wavelengths give RMSEP equal to or lower than the full spectrum with fewer variables.
- **H4 (RQ3b):** At least one non-linear model (SVR or RF) achieves lower RMSEP than the PLSR baseline.

- **H5 (RQ4):** Field-measured predictions take under 60 s per sample with RMSEP no more than 1.5× the laboratory RMSEP.

Development methodology: iterative and incremental (a lightweight Agile/CRISP-DM hybrid). Each iteration adds one pipeline component (e.g. a preprocessing method), evaluates it with cross-validation, and records the result before moving on.

Implementation process

1. **Protocol design.** Write the standard operating procedure (SOP) for sampling, scanning and drying; review with the supervisor and, if possible, a tea technologist.
2. **Pilot study (10–15 samples).** Check spectrometer settings, scan repeatability, sample-holder geometry and drying time to constant mass.
3. **Main data collection.** Collect, scan and dry the full sample set in batches, logging ambient conditions.
4. **Data cleaning and QC.** Inspect spectra, remove noisy edges of the range, run PCA outlier checks, check reference triplicates.
5. **Data splitting.** Lock the test set before any modelling.
6. **Preprocessing experiments.** Fix PLSR as the model; compare preprocessing combinations by RMSECV.
7. **Wavelength-selection experiments.** Apply CARS and a comparison method to the best preprocessing; repeat CARS runs to check stability.
8. **Model experiments.** Train PLSR, SVR, RF (and optionally ANN/1D-CNN) on the best one to three preprocessing/selection settings; tune by cross-validation.
9. **Final evaluation.** Evaluate the shortlisted pipelines once on the locked test set.
10. **Prototype.** Package the chosen pipeline into the CLI/Streamlit tool.
11. **Field feasibility test.** Use the prototype at a factory, store or non-lab setting; oven-check a subset.
12. **Analysis and write-up.** Interpret results against the RQs and hypotheses; document limitations.

Data sources and collection

- **Samples:** black tea (made tea, mainly BOP, BOPF, Pekoe, Dust and similar grades) from several Sri Lankan sources, ideally covering low-, mid- and high-grown regions. Targets: at least 120 samples, with a minimum of 80.
- **Moisture range:** made tea alone sits in a narrow band (about 2–7%). To widen the range for modelling, some samples will come from late drying stages where accessible, and others will be conditioned in controlled humidity (e.g. sealed containers over saturated salt solutions) or briefly dried, giving roughly 2–15% moisture.

- **Handling:** samples sealed in airtight bags, brought to room temperature before scanning, and scanned and weighed for drying within a short, fixed time window to prevent moisture exchange.
- **Spectra:** dark and white-reference before each session and every 30 minutes; three scans per sample with repacking; sample presented in a fixed-depth holder or glass cup.
- **Reference moisture:** about 3–5 g per replicate, dried at 103 ± 2 °C to constant mass (difference < 0.001 g between weighings), cooled in a desiccator; moisture (wet basis) = (initial mass − dry mass) / initial mass × 100; triplicate, mean used.
- **Metadata:** source, grade, region, processing stage, collection date, ambient temperature and humidity, operator.
- **Secondary data:** published results from the literature matrix for benchmarking only; no external dataset is merged into training.

Dataset suitability criteria. No suitable public dataset exists (reviewed datasets are Chinese, Indian or Spanish teas on other instruments, mostly available only on request), so primary data will be collected. It is checked against:

Criterion	Question	Plan
Relevance	Does it represent the problem and target?	Sri Lankan black tea with oven moisture for every spectrum
Features	Are the needed variables present?	Full spectrum, plus source, grade, stage, ambient temperature and humidity
Sample size	Enough for the models planned?	Target 120 (minimum 80); neural models only above about 150
Target distribution	Is moisture spread or bunched?	Check histogram after the pilot; widen with drying-stage and conditioned samples
Quality	Noise, outliers, label error?	Triplicate reference, PCA outlier checks, control-sample drift checks
Population / context	What does it represent and not represent?	Selected estates, grades and one season; not all regions, cultivars or years
Licence / ethics	Is use permitted? Any sensitive data?	Supplier consent for named sources; no personal data

Criterion	Question	Plan
Validation role	Training, internal test or external test?	Calibration set, locked internal test set, and source-held-out test
Literature data	Can published data be reused?	Used for benchmarking only; not merged, because instruments and teas differ

Key methods and procedures

Preprocessing

- **SNV:** centre and scale each spectrum by its own mean and standard deviation to remove scatter effects from particle size and packing.
- **MSC:** regress each spectrum against the mean calibration spectrum and correct offset and slope.
- **Savitzky-Golay:** fit a local polynomial in a moving window (e.g. window 7–21 points, polynomial order 2) for smoothing and first/second derivatives.
- **Combinations:** SNV + SG, MSC + SG, SG-derivative + SNV, each followed by mean-centring.

Wavelength selection

- **CARS:** repeated Monte Carlo sampling with PLS regression coefficients; wavelengths with small coefficients are removed in an exponentially decreasing schedule; the subset with the lowest RMSECV is kept. Run 20–50 times to check how stable the selected bands are.
- **Comparison:** SPA or interval PLS; VIP scores for interpretation.
- **Interpretation check:** selected wavelengths are compared with known water bands (about 970, 1,200 and 1,450 nm).

Models

- **PLSR:** number of latent variables chosen by the lowest RMSECV (or a one-standard-error rule to avoid overfitting).
- **SVR:** RBF kernel; C, γ and ε tuned by grid or Bayesian search within cross-validation.
- **Random Forest:** number of trees, depth and features per split tuned by cross-validation.
- **ANN / 1D-CNN (optional):** small network with early stopping, only if sample count supports it.

Validation

- Split about 70–75% calibration and 25–30% test, using Kennard-Stone (or SPXY) on the calibration side to cover the space, plus a check that the test set spans the moisture range.
- Where at least three sources are available, an additional **leave-one-source-out** test is run to measure generalisation to an unseen estate.
- Inner 10-fold (or repeated) cross-validation on calibration data for all tuning; the test set is used once.

Reproducibility and quality controls

- **Written SOPs** for scanning and drying; same operator and instrument settings throughout.
- **Instrument checks:** white and dark references; a fixed control sample scanned at the start of each session to detect drift.
- **Reference QC:** triplicate drying; samples with replicate SD > 0.2% re-run or excluded with a logged reason.
- **Randomised scanning order** to avoid confounding moisture with time of day.
- **No data leakage:** preprocessing parameters, wavelength selection and tuning are fitted inside cross-validation on calibration data only (scikit-learn pipelines).
- **Fixed random seeds**, pinned library versions (requirements.txt) and Git commits for every reported result.
- **Experiment log** (MLflow or CSV) recording pipeline, parameters and metrics for every run.
- **Data versioning:** raw data kept read-only; cleaned data produced by scripts.

Implementation phases

Phase	Activities	Deliverable
P1. Research and planning	Literature review, survey matrix, gap analysis, RQs and objectives, project proposal, ethics form	Project proposal (PPRS) and literature survey matrix
P2. Protocol and procurement	Confirm spectrometer, oven and balance access; write SOPs; arrange sample suppliers; pilot study	SOP document, pilot report, equipment confirmation
P3. Data collection	Scan and oven-dry all samples; log metadata and ambient conditions; QC checks	Paired dataset v1 with data dictionary

Phase	Activities	Deliverable
P4. Data preparation and baseline	Cleaning, outlier detection, data split, PLSR on raw spectra	Clean dataset v2, locked test set, baseline results
P5. Preprocessing and wavelength selection	SNV/MSC/SG/derivative experiments; CARS and comparison method	Preprocessing and wavelength-selection comparison tables (RQ2, RQ3)
P6. Model development	PLSR, SVR, RF, optional ANN/1D-CNN; tuning; shortlisting	Trained models and model comparison report
P7. Evaluation and prototype	Final test-set evaluation; build CLI/Streamlit prototype	Evaluation report (RQ1) and working prototype
P8. Field feasibility	Field use of prototype; oven-check subset; timing and usability notes	Field feasibility report (RQ4)
P9. Write-up and defence	Thesis chapters, viva preparation, code and data packaging	Final thesis, code repository, presentation

Objective → method → evidence → decision

Objective	Method	Evidence	Decision it supports
O1 Literature investigation	Structured review and thematic synthesis	Survey matrix, themes, gap evidence test	Which parameter, sensor, methods and validation design are justified
O2 Build dataset	SOP-based acquisition and oven reference	Paired dataset, QC report	Is the data good enough to model?
O3 Preprocessing	Factorial comparison under fixed PLSR	RMSECV table, ablation	Which preprocessing goes forward
O4 Wavelength selection	CARS vs SPA/iPLS vs full spectrum	RMSECV, variable count, band stability	Whether a reduced band set is viable for low-cost devices
O5 Model comparison	PLSR, SVR, RF on identical splits	RMSECV, RMSEP, paired tests	Whether non-linear models justify their complexity

Objective	Method	Evidence	Decision it supports
O6 Held-out evaluation	Single locked-test run; leave-one-source-out	RMSEP, R^2p, RPD, bias, residuals	Can NIR estimate moisture for screening?
O7 Field feasibility	Prototype trial with oven subset	Time, repeatability, field RMSEP, SUS	Is the approach practical outside the lab?

Methodology selection matrix

Objective / question	Evidence needed	Candidate methods	Why suitable	Limitations	Final choice
RQ1 accuracy vs reference	Prediction error on unseen samples	Random split; Kennard-Stone split; leave-one-source-out	KS covers spectral space; LOSO tests transfer	Small n gives wide error bars	KS locked test + LOSO where sources allow
RQ2 preprocessing effect	Error change vs raw spectra	One-at-a-time; full factorial grid	Grid shows interactions	More runs, risk of over-searching	Grid inside CV, test set untouched
RQ3 selection and models	Error and complexity for each option	CARS, SPA, iPLS, GA; PLSR, SVR, RF, ANN	Established and available in Python	CARS is stochastic	CARS (repeated) + SPA/iPLS; PLSR, SVR, RF, ANN optional
RQ4 feasibility	Time, robustness, field accuracy, usability	Lab simulation; field trial; user interviews	Field trial gives real conditions	Small field sample	Field trial + SUS + oven subset

Method-selection checklist

Check	Result
Does each method help achieve an objective?	Yes: see the objective → method table
Does it produce evidence for an RQ?	Yes: every method maps to RQ1–RQ4
Is it valid for spectra and this problem?	Yes: all methods are standard in NIR chemometrics (Qiu 2026)

Check	Result
Can it run with the data, time and resources?	Yes, if at least 80 samples are collected; ANN is optional
Can it be reproduced?	Yes: SOPs, fixed seeds, pinned versions, logged runs
Does it add ethical, legal or EDI risk?	Low: no personal data; consent for named suppliers
Is there a simpler method that answers the same question?	PLSR is kept as the simple answer; complex models must beat it to be preferred

9. Testing and Evaluation

Every pipeline is judged on the same locked test set against oven-drying, and the headline result is RMSEP, R^2p and RPD on samples the model never saw.

Evaluation questions

- How close are NIR moisture predictions to oven-drying values on unseen samples? (RQ1)
- How much do SNV, MSC, Savitzky-Golay and derivatives change accuracy compared with raw spectra? (RQ2)
- Does CARS (or another selection method) improve accuracy or keep it while using fewer wavelengths? (RQ3)
- Do SVR, RF or a neural model beat the PLSR baseline, and is the difference meaningful? (RQ3)
- Does accuracy hold for samples from a source not used in training? (Gap 3)
- How fast, repeatable and usable is the method outside the laboratory? (RQ4)

Baseline or comparison approach

- **Reference baseline:** oven-drying to constant mass (ground truth).
- **Model baseline:** PLSR on raw (mean-centred) full spectra.
- **Naïve baseline:** predicting the calibration-set mean for every sample. Any useful model must beat it clearly; this also gives the denominator for RPD.
- **Literature benchmark:** results reported by Zou et al. (2022), Lan et al. (2022) and Andasuryani et al. (2024), used as context only because samples and instruments differ.

- **Ablation:** remove one component at a time (preprocessing, selection, non-linear model) from the best pipeline to show what each contributes.

Metrics

$$\mathrm{RMSEP} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(\hat{y}_i - y_i)^2}$$

$$R^2 = 1 - \frac{\sum_{i}(\hat{y}_i - y_i)^2}{\sum_{i}(y_i - \bar{y})^2}$$

$$\mathrm{RPD} = \frac{SD_{\text{reference, test}}}{\mathrm{RMSEP}} \qquad \mathrm{Bias} = \frac{1}{n}\sum_{i=1}^{n}(\hat{y}_i - y_i)$$

RMSEC and RMSECV are reported alongside RMSEP to show overfitting (a large gap between calibration and prediction error is a warning sign). Typical interpretation of RPD: below 1.5 unusable, 1.5–2.0 rough screening, 2.0–2.5 approximate quantification, 2.5–3.0 good, above 3.0 excellent.

Why each metric is used:

Metric / evidence	What it tells us	Objective / requirement
RMSEP	Typical prediction error in % moisture on unseen samples, in the units an operator uses	O6, NFR01
R^2p	Share of moisture variation explained; comparable across studies	O6, NFR01
RPD	Error relative to the spread of the data; guards against a high R^2 on a narrow range	O6, NFR01
Bias	Systematic over- or under-prediction, important for sale weight	O6, O7
RMSEC vs RMSECV vs RMSEP gap	Overfitting and calibration-to-validation drop	O5, O6
Number of selected wavelengths	Model simplicity and suitability for cheaper sensors	O4
CARS selection frequency over repeated runs	Stability of selected bands	O4

Metric / evidence	What it tells us	Objective / requirement
Paired Wilcoxon test on absolute errors	Whether a model's improvement over PLSR is more than chance	O5
Per-source RMSEP (LOSO)	Generalisation to an unseen estate or factory	O6 (Gap 3)
Repeat-scan SD	Measurement repeatability	NFR03
Time per measurement	Practical speed vs oven-drying	O7, NFR02
SUS score	Usability for operators	O7, NFR06
Training / prediction time and CPU use	Computational cost (sustainability)	NFR05

Accuracy (classification-style "% correct") is not used: this is a regression problem, and a single accuracy figure would hide the size of errors.

If the proposed pipelines do not beat PLSR: that is a valid finding. It would show that a simple linear model is enough for tea moisture at this sample size, which is useful for low-cost deployment. The report will state this directly rather than present a weaker model as better.

Success criteria

Level	R^2p	RPD	RMSEP (made tea)	Interpretation
Minimum (feasible for screening)	⩾ 0.80	⩾ 2.0	⩽ 1.0%	Can separate clearly wet / acceptable / over-dried lots
Target	⩾ 0.90	⩾ 2.5	⩽ 0.5%	Useful preliminary measurement before lab confirmation
Stretch	⩾ 0.95	⩾ 3.0	⩽ 0.3%	Close to laboratory repeatability

In addition: time per measurement < 60 s; prediction repeatability SD ⩽ 0.2%; field RMSEP ⩽ 1.5× laboratory RMSEP.

Evaluation procedure

1. Lock the test set before modelling and record its hash.

2. Run all preprocessing × selection × model combinations with nested or repeated cross-validation on the calibration set.
3. Shortlist the best three pipelines by RMSECV (and model simplicity).
4. Evaluate the shortlist once on the test set; report every metric, not only the best one.
5. Compare the best model with PLSR using paired tests on absolute errors (Wilcoxon signed-rank) and, where suitable, a randomisation test on RMSEP difference.
6. Run leave-one-source-out validation when sample sources allow.
7. Analyse residuals by moisture level, grade, source and ambient humidity.
8. Field test: take 20–30 measurements with the prototype in a non-laboratory setting; oven-dry a subset of 10–15; record time per measurement, repeatability, ambient conditions and user feedback (short SUS questionnaire).
9. Summarise findings against hypotheses H1–H5.

Limitations of the evaluation

- Sample numbers (about 80–150) limit how complex the models can be and how precise the metrics are.
- One spectrometer; results may not transfer to other instruments without calibration transfer.
- One collection season; seasonal and year-to-year variation is not tested.
- Conditioning samples to widen the moisture range may create moisture distributions that differ from naturally produced tea.
- The oven method measures all volatiles lost at 103 °C, not only water, so the reference itself has small systematic error.
- Field testing is small-scale and can only indicate feasibility, not prove industrial readiness.

Traceability: objectives and requirements to evaluation

Objective / requirement	Method	Metric / evidence	Acceptance criteria
O1 / RQ1–RQ4 grounding	Structured literature review and survey matrix	Number of sources reviewed; gap matrix linking gaps to RQs	≥ 25 relevant sources; every RQ linked to at least one gap

Objective / requirement	Method	Metric / evidence	Acceptance criteria
O2 / FR01–FR05	Data collection with SOPs and QC	Sample count, moisture range, reference replicate SD	⩾ 80 samples (target 120); replicate SD ⩽ 0.2%; complete metadata
O3 / FR06 / RQ2	Preprocessing comparison with fixed PLSR and CV	RMSECV and R^2cv per preprocessing	Best preprocessing lowers RMSECV vs raw; differences reported for all methods
O4 / FR09 / RQ3	CARS vs full spectrum vs comparison method	RMSECV, number of variables, selection stability, band interpretation	Selected subset matches or beats full spectrum with fewer variables
O5 / FR10–FR12 / RQ3	Model comparison on same splits	RMSECV, RMSEP, paired significance test	Best model identified; difference vs PLSR quantified
O6 / FR13 / NFR01 / RQ1	Single evaluation on locked test set	RMSEP, R^2p, RPD, bias, residual plots	Meets at least the minimum level; target $R^2p \geqslant 0.90$, RPD ⩾ 2.5
Generalisation (Gap 3)	Leave-one-source-out validation	RMSEP per held-out source	Degradation reported; within 1.5× of test RMSEP considered robust
O7 / FR16–FR18 / NFR02–NFR03 / RQ4	Field test with prototype and oven subset	Time per sample, repeatability SD, field RMSEP, bias	< 60 s per sample; SD ⩽ 0.2%; field RMSEP ⩽ 1.5× lab RMSEP
NFR06 usability	Short user trial with SUS questionnaire	SUS score, observed errors	SUS ⩾ 68
NFR04 reproducibility	Re-run pipeline from repository on a clean machine	Identical metrics from saved seeds and data	Results reproduce to reported precision

10. Responsible Practice and Sustainability (LESP, EDI and BCS Code of Conduct)

The main responsibility risk is overclaiming: a screening model must never be presented or used as a certified laboratory result.

Mapping to the BCS Code of Conduct

BCS principle	Project responsibility	Tea Sense actions
Public Interest	Protect health, safety, privacy and the environment; avoid discrimination; promote equal access	Predictions labelled as screening only so unsafe (mouldy) tea is not released on a model's word; no personal data; reagent-free method; report accuracy per source so smaller producers are not disadvantaged
Professional Competence and Integrity	Work within competence; do not overstate expertise; follow law; accept criticism	Use methods I can explain and justify; seek supervisor and tea-technologist review of the protocol; report poor or negative results honestly; learn the data-protection and lab-safety rules that apply
Duty to Relevant Authority	Due care; protect confidential information; do not misrepresent performance	Follow IIT/UoW ethics process; keep supplier data confidential; report RMSEC and RMSEP side by side; disclose leakage risks and failure cases
Duty to the Profession	Uphold standards and good practice	Reproducible code, correct citation, version control, transparent AI-use declaration, publish protocol and code for others to check

Legal considerations

- **Data protection:** the study collects no personal data about tea samples. Usability feedback is anonymous; any personal data (consent forms) is handled under the Sri Lanka Personal Data Protection Act No. 9 of 2022 and university policy.
- **Confidentiality and commercial agreements:** supplier names and lot details are coded; written consent is obtained before naming any estate or factory.
- **Intellectual property and licences:** only open-source libraries with permissive licences (BSD, MIT, Apache) are used; spectrometer vendor software is used under its licence; all papers are cited.
- **Sector rules:** the project does not claim compliance with ISO 3720, ISO 1573 or buyer specifications; it uses ISO 1573-style drying only as a reference method.
- **Health and safety:** lab work follows the institution's safety rules for ovens, glassware and desiccants.

Ethical considerations

- **Potential harm:** an under-predicted moisture value could lead to mouldy tea being stored or shipped. The prototype therefore shows an out-of-range warning and the words "screening estimate, confirm in lab".
- **Human participants:** only for brief usability feedback in the field test; participants get an information sheet, give consent, can withdraw, and are not identified.
- **Over-reliance on automation:** the system supports staff decisions and does not replace laboratory confirmation.
- **Transparency:** limitations, error ranges and the conditions tested are published with the results.

Social considerations

- **Who benefits:** factories, brokers and exporters gain faster checks; laboratories gain fewer routine samples.
- **Who could be disadvantaged:** smaller estates if the model works poorly on their tea or if they cannot afford a spectrometer. Per-source reporting makes this visible.
- **Jobs:** the tool reduces routine testing load but keeps trained staff in control of confirmation and decisions.
- **Misuse outside intended context:** e.g. using it on green tea or another instrument. The README and interface state the validated scope.
- **Understandability:** output is a single % value with a clear status, not raw spectra.

Professional considerations

- Report only what has been done; no fabricated data, results or feedback.
- Report negative and weak results, including any failure to beat PLSR.
- Attribute third-party code, datasets and ideas.
- Keep reproducible records: SOPs, Git history, experiment logs, data versions.
- Use generative AI responsibly (see below).

Equality, Diversity and Inclusion (EDI)

EDI question	Tea Sense analysis
Representation	The dataset may over-represent accessible or larger producers and certain elevations (low-, mid- or high-grown). Sources and grades will be recorded and reported; gaps will be stated
Performance differences	Per-source and per-grade RMSEP will show whether the model works worse for some producers
Accessibility	Interface uses large numerals, text plus colour (not colour alone), few steps; Sinhala and Tamil labels are proposed as future work for factory operators
Bias amplification	A model trained on mostly one region could systematically misjudge others' tea, affecting their sale weight; LOSO testing checks this
Claims	No claim of validity for all Sri Lankan teas, cultivars, seasons or instruments without evidence
Equal access	Use of a low-cost portable sensor and free software lowers the barrier for smaller producers

Sustainability and SDGs

Area	Tea Sense
Environmental	NIR needs no reagents and destroys no sample; fewer oven runs save energy; dried tea from reference tests is composted

Area	Tea Sense
Computational	Small models (PLSR, SVR, RF) train in seconds on a laptop; deep models are only trained if justified; tuning grids are bounded
Operational	Maintainable, documented code and SOPs so the pipeline can be re-run and re-trained
Social	Designed around affordable hardware so smaller producers are not excluded
SDG 9 (Industry, Innovation and Infrastructure)	Brings low-cost sensing and data analysis to tea processing
SDG 12 (Responsible Consumption and Production)	Reduces waste from destructive testing and spoiled, mis-dried tea
SDG 8 (Decent Work and Economic Growth)	Supports quality and value in an export industry that employs many people in Sri Lanka

Responsible use of generative AI

- **Permitted use:** language refinement, structuring and clarity, as declared in the proposal's AI disclosure.
- **Prohibited use:** generating data, results, citations or analysis presented as the author's own findings.
- **Verification:** every reference is checked against the original paper; every number comes from the dataset or a cited source.
- **Confidentiality:** no supplier data or unpublished results are pasted into AI tools.
- **Register:** an AI-use register (date, tool, purpose, what was kept) is kept in Appendix L.

LESP + EDI + sustainability matrix

Issue	Evidence / context	Risk / impact	BCS principle	Mitigation / safeguard	How verified
Misuse / over-reliance	Screening model with known error	Mouldy or over-dried tea released; disputes	Public Interest; Duty to Relevant Authority	"Screening estimate" label; out-of-range warning; lab confirmation advised	UI review by supervisor; field users' understanding checked

Issue	Evidence / context	Risk / impact	BCS principle	Mitigation / safeguard	How verified
Performance misrepresentation	Calibration often overstates accuracy (Revilla 2024)	Overclaimed results	Professional Integrity; Duty to Relevant Authority	Locked test set; report RMSEC and RMSEP; report negatives	Experiment log; reproducibility re-run
Supplier confidentiality	Commercial samples	Reputational harm to suppliers	Duty to Relevant Authority	Coded sources; consent before naming	Data dictionary review; consent forms
Privacy	Usability feedback, consent forms	Personal data exposure	Public Interest	Anonymous feedback; minimal personal data; secure storage	Ethics approval; dataset audit
Bias / EDI	Uneven sampling of regions and producers	Worse accuracy for some producers	Public Interest	Diverse sourcing; per-source metrics; scope statements	Per-source residual analysis
Licensing / IP	Open-source and vendor software	Licence breach	Professional Integrity	Licence list; citations	README licence section
Lab safety	Oven at 103 °C, glassware	Injury	Public Interest	Induction, PPE, risk assessment	Signed risk assessment
Compute / sustainability	Model tuning and oven energy	Unnecessary energy use	Public Interest	Small models; bounded tuning; batched oven runs	Run times and oven-run counts logged
Generative AI	Writing support	Fabricated citations or content	Duty to the Profession	Verify all citations; AI-use register	Supervisor review; Appendix L

11. Project Timeline and Milestones

The plan runs from September 2026 to the end of May 2027, with data collection as the longest and riskiest phase; all dates are indicative until checked against official IIT/UoW deadlines.

Work breakdown structure

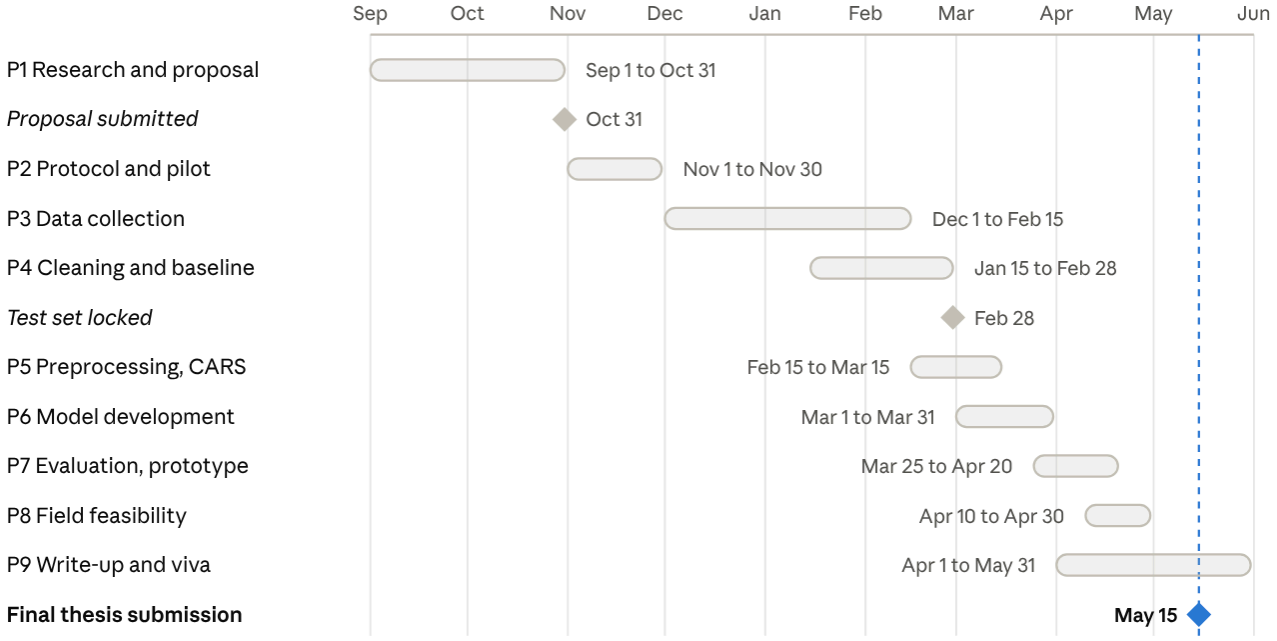
The work is broken into tasks with clear outputs before it is scheduled.

Workstream	Tasks	Output
Research	Finish thematic review; verify gap; justify methodology; complete [verify] references	Literature chapter, gap matrix
Ethics and approvals	Ethics form; lab risk assessment; supplier consent; participant information sheet	Approved ethics application
Protocol	Scanning SOP; drying SOP; sample-holder design; control-sample plan	SOP document
Pilot	Scan and dry 10–15 samples; check repeatability and drying time	Pilot report, revised SOPs
Data collection	Source samples; condition for moisture range; scan; oven-dry in triplicate; log metadata	Paired dataset v1
Data preparation	Clean spectra; outlier checks; data dictionary; lock test set	Dataset v2, locked test set hash
Baseline	PLSR on raw spectra; naïve mean model	Baseline metrics
Preprocessing	SNV, MSC, SG, derivatives and combinations under CV	Preprocessing comparison table
Wavelength selection	Repeated CARS; SPA or iPLS; band interpretation	Selection comparison, stability plot
Candidate models	SVR, RF (optional ANN); tuning; logging	Tuned models, run log
Evaluation	Locked-test run; LOSO; paired tests; residual analysis	Evaluation report
Artefact	Save pipeline; CLI and Streamlit prototype; out-of-range check; testing	Working prototype
Field feasibility	Field session; oven subset; timing; SUS	Field feasibility report

Workstream	Tasks	Output
Reporting	Methodology, results, discussion, limitations, conclusion, references, proofreading	Thesis draft and final
Assessment preparation	PP slides; interim demo; final demo; viva rehearsal	Slides, demo, Q&A notes

Data collection is the longest phase; final thesis due mid-May 2027

Tea Sense plan by month, Sep 2026 to May 2027 (indicative dates)



Tea Sense timeline · 9 phases, 3 milestones

Phases overlap on purpose: baseline modelling starts on early batches while collection continues, and write-up starts during evaluation.

Phase / task	Milestone / deliverable	Dependency
P1 Research and proposal	Literature survey matrix; project proposal submitted	Supervisor feedback
P2 Protocol and pilot	SOPs; pilot report; spectrometer and oven access confirmed	P1; equipment access; ethics approval
P3 Data collection	Paired dataset v1 (target 120 samples)	P2; sample suppliers; lab time

Phase / task	Milestone / deliverable	Dependency
P4 Cleaning and baseline	Clean dataset v2; locked test set; PLSR baseline	First P3 batches; complete dataset for test lock
P5 Preprocessing and wavelength selection	Comparison tables for RQ2 and RQ3	P4
P6 Model development	Tuned PLSR, SVR, RF (optional ANN)	P5
P7 Evaluation and prototype	Test-set evaluation report (RQ1); working prototype	P6
P8 Field feasibility	Field feasibility report (RQ4)	P7 prototype; site access; oven for subset
P9 Write-up and viva	Final thesis, code repository, presentation	P5–P8 results

12. Risk Management

The two risks that could sink the project are losing access to a spectrometer and collecting too few, too similar samples; both have early checks and fallbacks.

Risk	Likelihood	Impact	Mitigation	Contingency
R1. NIR spectrometer unavailable or delayed	Medium	High	Confirm access in P2; book instrument time early; identify a second source (university lab, partner institute)	Use a benchtop NIR at a partner lab; reduce field test scope
R2. Too few samples or too little moisture variation	Medium	High	Contact several suppliers early; include late drying stages; controlled humidity conditioning to widen range	Accept minimum of 80 samples; report reduced statistical power; drop neural models

Risk	Likelihood	Impact	Mitigation	Contingency
R3. Inconsistent reference moisture values	Medium	High	Written SOP; triplicate drying; desiccator cooling; calibrated balance; pilot study	Re-run high-variance samples; exclude with logged reason; cross-check with halogen analyser
R4. Moisture changes between scanning and drying	Medium	Medium	Airtight bags; fixed short time window; scan and weigh in the same room	Re-scan and re-weigh; record time gap and test its effect
R5. Poor model performance (R^2p < 0.80)	Medium	Medium	Systematic preprocessing / selection / model search; check outliers and reference quality	Report as a valid negative or partial result; analyse causes (range, noise, instrument)
R6. Overfitting / data leakage gives misleading results	Medium	High	Locked test set; scikit-learn pipelines inside CV; one-shot test evaluation	Re-split with a fresh lock if leakage is found; disclose it
R7. Field conditions (humidity, light, temperature) degrade predictions	High	Medium	Log ambient conditions; scan in a shaded holder; allow instrument warm-up	Treat as a feasibility finding; propose correction methods as future work
R8. Instrument drift between sessions	Medium	Medium	White/dark references; control sample each session; randomised order	Apply drift correction using control sample; flag affected batches
R9. Timeline slippage (coursework load, lab access)	Medium	Medium	Gantt chart with buffer; weekly supervisor check-ins; prioritise Must requirements	Drop Could items (ANN, GUI, extra parameters); reduce field test size
R10. Data loss or file corruption	Low	High	Git for code; two cloud backups for data; raw data read-only	Restore from backup; re-export from instrument memory

Risk	Likelihood	Impact	Mitigation	Contingency
R11. Limited skills in chemometrics	Medium	Medium	Early learning plan (online courses, papers, supervisor guidance); start from established libraries	Seek help from a domain expert; simplify to PLSR-centred study
R12. Ethics approval or supplier consent delayed	Low	Medium	Submit ethics form early; use retail samples that need no consent	Start with retail samples; add supplier samples later

Additional project-management risks:

Risk	Likelihood	Impact	Mitigation	Contingency
R13. Scope creep (adding polyphenols, more tea types, a mobile app)	High	High	MoSCoW priorities; moisture-only scope agreed with supervisor	Move extras to future work
R14. Narrow moisture range in made tea makes R^2 look poor even when error is small	High	Medium	Widen range with drying-stage and conditioned samples; report RMSEP and RPD, not R^2 alone	Present results for made-tea range separately with RMSEP
R15. Supplier withdraws or cannot provide samples	Medium	Medium	Agree quantities early; keep several suppliers; retail samples as backup	Use retail black tea of several grades

13. Expected Outcomes and Deliverables

The project will deliver a dataset, a ranked comparison of pipelines, a working prototype and a clear feasibility verdict for rapid NIR moisture estimation in black tea.

Expected outputs

Deliverable	Expected output	Linked objective
Literature review	Critical review of spectroscopy and AI methods for tea quality and moisture; survey and gap matrices	O1
Sample and reference dataset	Documented black-tea samples with oven-drying reference moisture (triplicate)	O2
NIR spectral dataset	Spectra paired with reference values, metadata and data dictionary	O2
Preprocessing evaluation	Comparison of raw, SNV, MSC, SG and derivative combinations	O3
Wavelength-selection evaluation	CARS and comparison method results, with selected bands linked to water absorption	O4
Predictive models	Tuned PLSR, SVR, RF (and optional ANN/1D-CNN) with saved artefacts	O5
Model evaluation report	RMSEP, R^2p, RPD, bias, significance tests and residual analysis on the locked test set	O6
Prototype	CLI and/or Streamlit tool that predicts moisture from a new spectrum with out-of-range warning	O5, O7
Field feasibility report	Time per sample, repeatability, field accuracy, usability and limitations	O7
Code repository and SOPs	Version-controlled code, environment file, SOPs and README	O2–O7
Final thesis and viva	Evidence-based conclusion on feasibility, limitations and future work	O6, O7

Expected benefits

- **For industry:** a moisture estimate in under a minute instead of hours; less sample waste; earlier dryer adjustments; fewer samples sent to the lab.

- **For research:** an NIR moisture model for Sri Lankan black tea, which none of the reviewed studies provide, and a fair, reproducible comparison of preprocessing, selection and models.
- **For sustainability:** fewer oven runs and no chemical reagents for routine screening.
- **For future work:** a reusable pipeline that can be re-trained for polyphenols, theaflavins or other parameters.
- **For the researcher:** applied skills in spectroscopy, chemometrics, machine learning and experimental research.

How outcomes will be measured

- Accuracy against the success criteria in Section 9 (R^2p, RPD, RMSEP, bias).
- Speed: measured time per sample versus oven-drying time.
- Robustness: leave-one-source-out results and field RMSEP.
- Usability: SUS score and observed errors from field users.
- Reproducibility: re-running the repository produces the reported numbers.
- Completion of all Must requirements in Section 6.

Known limitations

- Results apply to the samples, cultivars, regions, season and spectrometer used.
- Made-tea moisture spans a narrow range; conditioning may make the training range wider than real production.
- Sample size limits model complexity and the precision of metrics.
- Field testing is small and indicates feasibility, not industrial readiness.
- Only moisture is targeted; other quality parameters are future work.
- The oven reference has its own error, which sets a floor on achievable RMSEP.

14. References and Supporting Material

References are in Harvard style and come from the literature survey matrix; items marked [verify] need their venue or DOI confirmed before submission.

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Standards and statistics to cite (add full details): ISO 1573 (tea: determination of loss in mass at 103 °C); ISO 3720 (black tea: definition and basic requirements); Sri Lanka Tea Board export statistics for 2025 (source of the 245,787 t figure); Saunders, Lewis and Thornhill, *Research Methods for Business Students* (Research Onion).

Supporting material (appendices)

Appendix	Content
A	Detailed search strategy and screening log (33 records screened, 5 excluded as off-domain)
B	Concept map: tea quality → moisture → NIR → preprocessing → selection → models → validation → feasibility
C	Literature survey matrix (from the research spreadsheet)
D	Research gap analysis matrix (Gaps 1–4 mapped to RQs and objectives)

Appendix	Content
E	Dataset comparison matrix (sample counts, ranges, instruments of reviewed studies) and data dictionary
F	Stakeholder onion, elicitation instruments (interview/questionnaire) and evidence
G	Full functional and non-functional requirements
H	Use cases (scan sample, record reference, train model, predict moisture) and activity/sequence diagrams
I	Detailed architecture and data-flow diagrams
J	Detailed Gantt chart, backlog and milestone plan
K	Full risk register and issue log
L	Generative AI use declaration and AI-use register
M	Preliminary proof-of-concept screenshots, test logs and sample outputs
N	Ethics clearance material, consent forms and data-management approvals

15. Final Review Checklist

- [x] ~~The project begins with a significant computing problem rather than a preferred tool (Sections 0, 2)~~
- [x] ~~The problem is supported by credible evidence (Section 2)~~
- [x] ~~The working title reflects the actual scope (Section 0; refined title proposed)~~
- [x] ~~The aim is singular and clear (Section 3)~~
- [x] ~~The objectives are measurable and together achieve the aim (Section 3)~~
- [x] ~~The literature review uses a documented search and screening process (Section 4)~~
- [x] ~~The literature is synthesised by theme, not paper by paper (Section 4)~~
- [x] ~~The gap is supported by several pieces of evidence (Section 4, gap evidence test)~~
- [x] ~~Existing and similar solutions are analysed (Section 4)~~
- [x] ~~Requirements are derived from evidence (Sections 3 and 6)~~
- [x] ~~Tool and technology alternatives are compared (Section 7)~~
- [x] ~~The methodology suits the type of research (Section 8)~~

- [x] ~~Each major method is justified and linked to an objective (Section 8)~~
- [x] ~~Data selection and limitations are understood (Section 8, dataset criteria)~~
- [x] ~~Baselines and metrics are planned before implementation (Section 9)~~
- [x] ~~LESP, EDI and sustainability are project-specific and mapped to BCS principles (Section 10)~~
- [x] ~~Safeguards are stated for significant risks (Sections 10 and 12)~~
- [x] ~~The schedule has tasks, milestones, dependencies and contingency (Sections 11 and 12)~~
- [ ] The ethics application matches the planned methods (Section 16: to submit)
- [ ] The slide deck presents a coherent research story (Section 17: plan ready, slides to build)
- [ ] Decisions can be defended without slides (Section 18: rehearse)
- [ ] Spectrometer model and lab access confirmed with supervisor
- [ ] Sample suppliers and target sample count confirmed
- [ ] Timeline dates checked against official IIT/UoW deadlines
- [ ] [verify] references completed and the export statistic sourced

16. Ethics Application and Supervisor Review

The ethics application will describe exactly the methods in this proposal, and must be approved before any data collection or field testing starts.

Ethics application plan

Step	Tea Sense action	When
1. First ethics-risk scan	Done in Section 10: no personal data in the dataset; small human-participant element (usability feedback); supplier confidentiality; lab safety	Before methods are fixed (P1)
2. Human participants	Field-test users recruited voluntarily from the host site; information sheet; written consent; right to withdraw; anonymous SUS responses	Before P8
3. Organisational data	Written permission from each supplier to collect samples and, if named, to be identified; samples coded S1, S2, …	Before P3

Step	Tea Sense action	When
4. Consistency	Methods in the ethics form match Sections 8–9 (scanning, oven-drying, ML, field test, SUS)	At submission
5. Approval	Supervisor signature and ethics approval obtained before the formal deadline	End of P2
6. Change control	If the design changes (e.g. new participant activity, new data type), seek guidance and update the approval	Throughout

Risk category expected: low. No vulnerable groups, no sensitive personal data, no deception, no hazardous chemicals beyond standard lab ovens and desiccants.

Supervisor review checklist

Question	Where answered
Can the problem be stated in one sentence?	Section 2: oven-drying is too slow for operational decisions and NIR has not been validated on Sri Lankan black tea in practical conditions
Is the literature synthesis strong enough?	Section 4 (themes, gap evidence test, decisions informed)
Are aim and objectives aligned with the problem?	Section 3 (alignment map)
Are requirements and choices derived from evidence?	Sections 6 and 7; Section 4 decisions table
Is the methodology appropriate?	Section 8 (methodology family choice)
Does every important objective have evidence-producing methods?	Section 8 (objective → method → evidence → decision)
Are evaluation and baselines defined before implementation?	Section 9
Are LESP, EDI and sustainability project-specific?	Section 10
Is the project feasible in the time available?	Sections 11 and 12

Question	Where answered
Can every major choice be defended without slides?	Section 18

17. Project Proposal Slide Deck Plan

The slides present the final research argument in 16 slides; detailed matrices from this document are kept as backup slides for Q&A.

Slide	Content	Source in this proposal	Emphasis
1	Title, student, supervisor, one-sentence focus: "Can a portable NIR scan estimate black-tea moisture fast and accurately enough for screening?"	Cover	Very brief
2	Problem context and significance: tea moisture, 245,787 t exports, quality and weight impact	Section 2	High
3	Evidence-based problem statement (context → limitation → consequence → need)	Section 2	High
4	Aim and seven objectives, linked to RQ1–RQ4	Section 3	High
5	Literature review method: databases, concept-block search, screening layers, 33 → 28 sources	Section 4	Brief
6	Thematic synthesis: preprocessing, selection, models, validation, portability	Section 4	High
7	Existing solutions and their limits (oven, analysers, meters, in-line NIR, research models)	Section 4	Medium
8	Gap and gap evidence test, with implication for the project	Section 4	High
9	Key requirements (Must items)	Section 6	Medium
10	Proposed solution: pipeline diagram	Section 5	High
11	Tools and alternatives: sensor, spectrometer, reference, models, stack	Section 7	High

Slide	Content	Source in this proposal	Emphasis
12	Methodology and why it fits (experimental, artefact as instrument, Research Onion)	Section 8	High
13	Data, methods and evaluation plan: dataset criteria, splits, metrics, baselines, success levels	Sections 8–9	High
14	LESP + EDI + sustainability mapped to BCS principles	Section 10	High
15	Timeline, milestones and top five risks	Sections 11–12	Medium
16	Expected contribution, feasibility and alignment summary	Sections 3 and 13	High
Backup	Survey matrix, full requirements, full risk register, objective–method matrix, LESP matrix	Sections 4, 6, 8, 10, 12	Q&A only

Design rules: one message per slide; diagrams for the pipeline, timeline and alignment; citations on the slide where evidence is used; readable fonts and high contrast; no full-size matrices on main slides.

18. Live Defence: Prepared Answers to Examiner Questions

Each answer follows one pattern: answer directly, give the evidence, admit the limitation, state the decision.

#	Likely question	Prepared answer
1	What exactly is the computing problem, and what proves it matters?	Turning noisy, high-dimensional NIR spectra into reliable moisture estimates that hold on unseen tea. It matters because oven-drying takes hours and moisture drives safety, quality and sale weight; the evidence is in Section 2.

#	Likely question	Prepared answer
2	How did the literature change your idea?	I started wanting to detect moisture, polyphenols, glyphosate and metals. Revilla (2024) and Qiu (2026) showed trace contaminants have no direct NIR signal and weak validation, so I narrowed to moisture.
3	Which three sources influenced you most?	Zou et al. (2022): closest design for black-tea moisture with a micro-NIR. Hazarika et al. (2024): CARS + PLSR on black tea with a portable unit. Qiu et al. (2026): universality and environment are unresolved, which shaped my validation plan.
4	What is the key limitation in existing work you address?	Single-cultivar, single-site validation. I add held-out and source-held-out testing on Sri Lankan black tea and a field check.
5	Why is the aim achievable in one year?	One parameter, one instrument class, 80–120 samples, standard open-source methods, and Could-priority items that can be dropped.
6	How does each objective produce evidence?	See the objective → method → evidence → decision table in Section 8.
7	Why these requirements?	Each traces to a gap or risk: e.g. FR08 (locked test set) comes from Revilla's calibration-validation drop; FR17 (out-of-range warning) comes from the misuse risk.
8	What alternatives did you reject?	THz (water blocks it, Huang 2024), LIBS (destructive), Raman (insensitive to water), HSI (cost), Karl Fischer (reagents, cost), MATLAB (licence). Reasons are in Section 7.
9	Why is the methodology appropriate?	My questions are about measurable prediction error under controlled factors, so a quantitative experiment fits; the prototype is only the instrument for RQ4.

#	Likely question	Prepared answer
10	What is the difference between methodology and development process?	Methodology is how evidence is produced (controlled experiments, locked test). Development process is how I build the pipeline (iterative CRISP-DM cycle).
11	Why are your methods valid for spectra?	SNV/MSC correct scatter from particle size; SG derivatives remove baseline shifts; PLSR handles many correlated wavelengths with few samples. These are standard (Qiu 2026).
12	What is your baseline and why is it fair?	PLSR on raw spectra plus a mean-only model. All candidates use the same splits, the same CV folds and the same tuning budget.
13	Why these metrics?	RMSEP is error in % moisture; RPD guards against a high R^2 on a narrow range; bias matters for sale weight. Classification accuracy would hide error size.
14	How do you prevent leakage?	Test set locked before modelling; preprocessing, CARS and tuning fitted inside CV on calibration data only, using scikit-learn pipelines; repeated scans of one sample never split across sets.
15	What if your models don't beat PLSR?	That is a valid result: it shows a simple linear model is enough at this sample size, which suits low-cost devices. I will report it as such.
16	Main legal, ethical, social and professional risks?	Misuse as a lab result, supplier confidentiality, lab safety, and overclaiming results. Mitigations are in Section 10.
17	How does the BCS Code shape your decisions?	Public Interest: screening label and warnings. Integrity: honest negatives. Relevant Authority: confidentiality and no misrepresentation. Profession: reproducible, cited work.

#	Likely question	Prepared answer
18	What EDI risks exist?	Uneven sampling could make the model worse for smaller or under-sampled producers. I report per-source error and state the validated scope.
19	Biggest project risk and contingency?	Spectrometer access. I confirm it in P2 and have a benchtop fallback; if samples are short, I accept 80 and drop neural models.
20	What is technically hardest?	Getting enough moisture variation and consistent reference values; small reference errors set a floor on achievable RMSEP. The pilot study and triplicate drying address this.

Backup evidence to bring: the literature survey matrix, pilot results (if available), the SOPs, the risk register and the alignment map.